

1.What does the len() function do in Python?

The len() function in Python returns the number of items in an object. It can be used on various data types, such as lists, strings, tuples, dictionaries, and more. The result of len() is the count of elements in the object.

2.Difference between Local and Global Variables in a Python Function

- **Local Variables**: A local variable is defined within a function and is only accessible within that function's scope. Once the function execution is complete, the local variable is no longer available.
- **Global Variables**: A global variable is defined outside any function and is accessible throughout the entire program. Global variables can be accessed inside functions, but if you want to modify a global variable inside a function, you must declare it using the global keyword.

```
x = 10 # This is a global variable
```

```
def example_function():  
    x = 5 # This is a local variable  
    print("Local x:", x)
```

```
example_function()  
print("Global x:", x)
```

- ☐ In this example, there is a global variable x set to 10. Inside example_function, a local variable x is set to 5. When example_function runs, it prints the value of the local x, which is 5. However, the global x outside the function remains 10. This demonstrates how Python differentiates between local and global variables based on scope.